# Analysis Note 2: Cosmic Muons

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## Thresholds Tables

| PMT | Threshold Voltage (mV) | Counts | Count Rate (counts/sec) |
|---|---|---|---|
| Bar 1 Left | 97.5 | $30433 \pm 174$ | $1014 \pm 5.8$ |
| Bar 1 Right | 85.8 | $15347 \pm 124$ | $512 \pm 4.1$ |
| Bar 2 Right | 76.8 | $17362 \pm 132$ | $578 \pm 4.4$ |
| Bar 2 Left | 120.5 | $17776 \pm 133$ | $592 \pm 4.4$ |

Table 1: Threshold voltages and count rates for PMT detectors left and right sides.

## 2-Fold Count Rates

| PMT | Bar 1 | Bar 2 |
|---|---|---|
| | Trial 1 | Trial 2 |
| Coincidence 1 (counts/sec) | $93.5 \pm 1.8$ | $97.5 \pm 1.8$ |
| Coincidence 2 (counts/sec) | $107.7 \pm 1.9$ | $106.9 \pm 1.9$ |

Table 2: 2-fold coincidence rates for the top and bottom PMT tubes.

# 4-Fold Count Rates

| All PMT's Sides | Trial 1 | Trial 2 | Trial 3 | Trial 4 |
|---|---|---|---|---|
| Count Rates (counts/sec) | $5.333 \pm 0.422$ | $5.30 \pm 0.42$ | $4.733 \pm 0.397$ | $5.10 \pm 0.41$ |

Table 3: 4-fold coincidence rates for both PMT's including the sides.

# Count Rate vs. Angular Distribution of $\theta$ and $\cos^2(\theta)$

| $\theta$ (deg) | $\theta$ (rad) | Counts | Count Rate (counts/sec) |
|---|---|---|---|
| 90 | 1.5708 | $106 \pm 10$ | 3.533±0.343 |
| 80 | 1.3963 | $135 \pm 12$ | 4.5±0.4 |
| 70 | 1.2217 | $115 \pm 11$ | 3.833±0.357 |
| 60 | 1.0472 | $87 \pm 9$ | 2.9 ±0.3 |
| 50 | 0.8727 | $59 \pm 8$ | $1.966 \pm 0.256$ |
| 40 | 0.6981 | $53 \pm 7$ | 1.766 ±0.242 |
| 30 | 0.5235 | $33 \pm 6$ | $1.1 \pm 0.2$ |
| 20 | 0.3491 | $17 \pm 4$ | 0.566 ±0.137 |
| 10 | 0.1745 | $12 \pm 3$ | 0.4 ±0.1 |
| 0 | 0.0000 | $11 \pm 3$ | 0.366 ±0.110 |

Table 4: Angular distribution parameters that demonstrate muon zenith angular dependence.

# Time Delay Table

| Peak | Total Count | Time Delay ($\mu$s) | Mean Bin | RMS |
|---|---|---|---|---|
| 1 | $235 \pm 15$ | 1.46 | $1480 \pm 0.041$ | $0.617 \pm 0.032$ |
| 2 | $295 \pm 17$ | 2.60 | $2647.55 \pm 0.14$ | $2.221 \pm 0.126$ |
| 3 | $266 \pm 16$ | 3.82 | $3810.930 \pm 0.598$ | $6.217 \pm 0.502$ |
| 4 | $255 \pm 16$ | 5.02 | $4967.68 \pm 1.16$ | $9.310 \pm 0.913$ |
| 5 | $202 \pm 14$ | 6.16 | $6063.310 \pm 0.786$ | $9.225 \pm 0.670$ |
| 6 | $124 \pm 11$ | 7.32 | $6238.06 \pm 1.45$ | $10.370 \pm 1.519$ |
| 7 | $257 \pm 16$ | 8.00 | $6776.16 \pm 0.82$ | $9.850 \pm 0.840$ |

Table 5: This is the raw data for time delay calibration. Each peak is gaussian and follows multiple minute trials. Information here is what eventually leads to muon half-life predictions

## Time Calibration Parameters

(1)

## Count Rate vs. Zenith Fit

(2)

## Count Rate vs. $\cos^2(\theta)$ Fit

(3)